

Development of an Online Marathon System using Acoustic AR

Yuki Konishi
Kyoto Sangyo University
Kyoto, Japan
i2386067@cc.kyoto-su.ac.jp

Panote Siriaraya
Kyoto Institute of Technology
Kyoto, Japan
spanote@kit.ac.jp

Da Li
Fukuoka University
Fukuoka, Japan
lida@fukuoka-u.ac.jp

Katsumi Tanaka
University of Fukuchiyama
Kyoto, Japan
tanaka-katsumi@fukuchiyama.ac.jp

Yukiko Kawai
Kyoto Sangyo University
Kyoto, Japan
kawai@cc.kyoto-su.ac.jp

Shinsuke Nakajima
Kyoto Sangyo University
Kyoto, Japan
nakajima@cc.kyoto-su.ac.jp

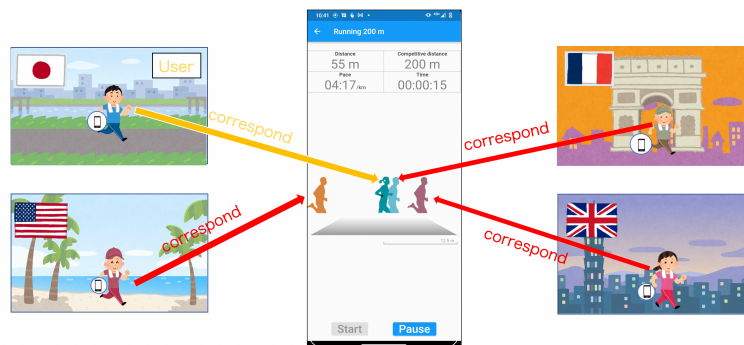


Figure 1: Overview of online marathon application.

ABSTRACT

In recent years, more and more people have taken up running to improve their health, but it is not easy to continue running. Therefore, we believe that the development of a running support system is important. We developed a running support system and application that allows users to run with a virtual runner created using their past running data in an acoustic augmented reality space. However, this system has a problem in that it only allows users to compete against past running records of themselves or their acquaintances. Therefore, in this paper, we propose an online marathon system that allows users to compete against an unspecified number of users, realize an online race where the distance and number of times can be selected arbitrarily, and investigate the effect of the number of competitors on user motivation.

CCS CONCEPTS

• **Information systems** → *Geographic information systems*; • **Applied computing** → *Health care information systems*; • **Human-centered computing** → *Smartphones*.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

MM '23, October 29–November 3, 2023, Ottawa, ON, Canada

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0108-5/23/10.

<https://doi.org/10.1145/3581783.3612672>

KEYWORDS

Running support system, acoustic augmented reality (AR), GPS

ACM Reference Format:

Yuki Konishi, Panote Siriaraya, Da Li, Katsumi Tanaka, Yukiko Kawai, and Shinsuke Nakajima. 2023. Development of an Online Marathon System using Acoustic AR. In *Proceedings of the 31st ACM International Conference on Multimedia (MM '23)*, October 29–November 3, 2023, Ottawa, ON, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3581783.3612672>

1 INTRODUCTION

Although the number of individuals striving to improve their health and physical fitness through running has been on the rise in recent years, sustaining motivation for running proves challenging, as research indicates that 40–65% of people quit exercising within 3–6 months of commencing. Therefore, we deem it crucial to develop a running support system. In our previous study, we developed a running support system that enables users to engage in running with a virtual runner in an acoustic augmented reality space, created using existing running data [5] [6]. However, the system had limitations that restricted racing to only past running records of oneself or acquaintances, and only for a single instance, thereby making it impossible to race against an unspecified number of users. It is necessary to develop a system that allows a large group of users to race against each other while providing the option to choose the distance and number of races. In this paper, we propose an online marathon system that facilitates a large-scale online race

while examining the effect of the number of players on the user's motivation (Fig. 1).

2 OVERVIEW OF AN ONLINE MARATHON SYSTEM USING ACOUSTIC AR

Fig. 2 presents a schematic diagram of the acoustic augmented reality (AR) running support system used in our online marathon system. A distinctive characteristic of this system is that users can compete against virtual runners while being immersed in the sound of their footsteps in the AR space. The speed and distance of the virtual runner and the user are calculated using GPS and updated every second. The distance traveled is calculated using the Harversine formula [4] based on latitude and longitude coordinates, and the results are added together. The user's speed is calculated in real-time based on the latitude and longitude of the nearby users, while the virtual runner's speed is calculated based on the average speed from past running data. The system uses the HRTF algorithm to create a 3D sound experience for the sound of footsteps [2]. The proposed online marathon app enables users to compete in marathon-style races with a large number of users, even those far away locations, by using a server. The recommended competitors during the race are selected based on the lap time and are typically users participating in the same marathon. In addition, to accommodate long-distance races, the system updates the recommended competitors every 100 meters.

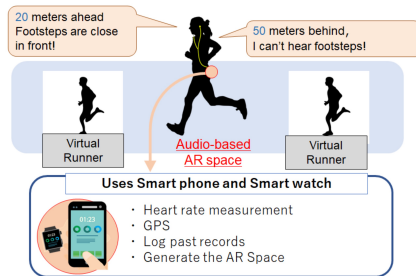


Figure 2: Schematic diagram of our acoustic AR running support system.

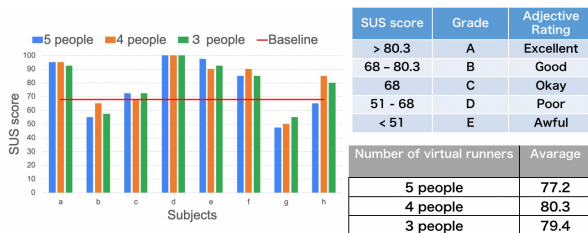


Figure 3: Experimental results (SUS questionnaire).

3 EXPERIMENT

We carried out an evaluation experiment to assess the usability of our developed online marathon application, as well as the physical

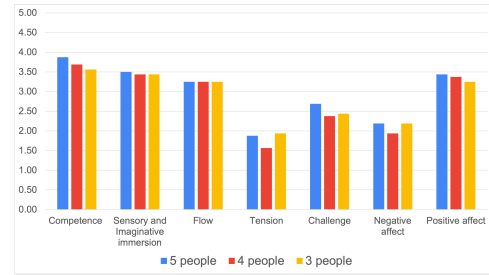


Figure 4: Experimental results (GEQ questionnaire).

and psychological effects on users of varying numbers of virtual runners presented during app use. Additionally, we conducted a survey to determine the appropriate number of virtual runners. The experiment lasted for three weeks, during which participants ran under three experimental conditions for one week each, with different numbers of virtual runners appearing (5 people, 4 people, and 3 people). We compared the speed and heart rate records on the first and last day of each condition and administered surveys to measure user usability (SUS) [1], user experience after gameplay (GEQ) [3], and a self-prepared questionnaire, all conducted after the final day of running.

4 CONCLUSION AND FUTURE WORK

The results of the evaluation experiments confirm that as for the usability of the application, as shown in the results of the sus questionnaire in Fig. 3, the system was highly rated in all conditions, indicating that a highly usable system had been developed. As for the results of the GEQ questionnaire, the tension level was low in all conditions, but generally, the game was better in the condition with five virtual runners (shown in Fig. 4). It was confirmed that the evaluation of the experience was higher in the condition with five virtual runners. Therefore, it is necessary to consider not only the number of virtual runners but also other factors affecting the recognition rate, such as voice and performance, in order to realize a competition with a moderate level of tension. Additional research will be conducted to enhance the recognition rate of virtual runners.

5 ACKNOWLEDGMENT

This work was supported in part by JSPS KAKENHI (20H04293 and 22H03700) and by a joint research project (M2301) of Kyoto Sangyo University.

REFERENCES

- [1] John Brooke et al. 1996. SUS-A quick and dirty usability scale. *Usability evaluation in industry* 189, 194 (1996), 4–7.
- [2] C. I. Cheng and G. H. Wakefield. 1999. Introduction to head-related transfer functions (HRTFs): Representations of HRTFs in time, frequency, and space. In *Audio Engineering Society Convention 107*. Audio Engineering Society.
- [3] Wijnand A IJsselstein, Yvonne AW De Kort, and Karolien Poels. 2013. The game experience questionnaire. (2013).
- [4] James Inman. 1849. *Navigation and nautical astronomy, for the use of British seamen*. F. & J. Rivington.
- [5] T. Kiri, M. Mittal, P. Shiriaraya, Y. Kawai, and S. Nakajima. 2020. Improving Motivation and Physical Impact Evaluation with an Acoustic AR Running Support System. *DEIM Forum 2020* (2020), 12–1.
- [6] Y. Konishi, P. Shiriaraya R. Arisawa, Da Li, Y. Kawai, and S. Nakajima. 2022. Verification of the Influence of Multiple Virtual Runners on Rival Recommendation for Acoustic AR Running Assistance System. *DEIM Forum 2022* (2022), A31–3.